

MATH6185 - Operational Research and Data Science Case Study 1

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1. Introduction

Healthcare appointment scheduling is a fundamental operational problem in service systems with fixed capacity and uncertain demand, in which healthcare providers must allocate limited appointment slots while balancing multiple objectives.

This study investigates a simplified healthcare appointment system with different patient groups.

Basic assumptions of this system are as follows:

- 1) All patients are either urgent (U) patients or routine (R) patients.
- 2) Arrivals of each patient type are modelled as independent Poisson processes with respective arrival rates.
- 3) Appointments are allocated immediately upon arrival.
- 4) Each working day contains 6 predefined appointment slots.
- 5) The long-run performance of scheduling policies is evaluated by Monte Carlo simulation.

The study evaluates all scheduling policies, with the results structured around the main questions:

- 1) The First Available Slot (FAS) policy is evaluated in Q2.
- 2) In Q3, FAS is extended by the workload-based delay mechanism.
- 3) In Q4, FAS is extended by on-call routine patients (OCR) and the workload-based reservation mechanism.

The following sections are structured as:

- 1) Section 2 reviews the relevant literature on operational problems defined in the coursework brief.
- 2) Section 3 describes the structure and design rationale of all scheduling policies.
- 3) Section 4 presents the simulation results of all policies and discusses the operational insights obtained.
- 4) Section 5 concludes the report.

2. Literature Review

Service systems with fixed capacity and multiple priority classes often face uncertain demand, motivating research into scheduling strategies to balance competing service requirements.

To address this challenge, the relevant literature serves different purposes:

- 1) Methodological reviews provide a broad understanding of simulation approaches and healthcare system characteristics.
- 2) Modelling studies provide technical foundations by proposing specific scheduling mechanisms and offering analytical insights.

2.1 Reviews of Healthcare Appointment Scheduling

- Review Scope

Cayirli and Veral (2003) provided a comprehensive review of outpatient appointment scheduling models in healthcare systems, where they categorised existing scheduling approaches according to system characteristics, scheduling rules and performance measures.

Gupta and Denton (2008) further reviewed mathematical models for healthcare appointment scheduling and classified existing approaches by demand uncertainty, service-time variability, patient behaviour and operational constraints.

Hulshof et al. (2012) extended the scope beyond appointment scheduling by reviewing broader healthcare operational planning problems that include resource management.

- Qualitative Insights

Across these reviews, appointment scheduling is consistently formulated as a demand–capacity balancing problem where limited service capacity must be allocated under uncertain future demand.

Moreover, the literature further suggests that increasing system utilisation generally improves resource efficiency but simultaneously reduces flexibility to respond to stochastic demand.

- Relevance to the Scheduling Policy

These reviews provide the conceptual foundation for this study and highlight the potential limitation of reallocation when the appointment system operates close to its capacity limit, as observed in **Section 4.3**.

2.2 Patrick and Puterman (2007)

- Essential Assumptions

Patrick and Puterman (2007) investigated similar flexible scheduling policies for diagnostic services, where they proposed a two-stage scheduling mechanism – during the initial scheduling stage, a proportion of appointment capacity was reserved for high-priority patients; as the appointment date approached, any unused reserved capacity could be reallocated to lower-priority call-in patients.

- Main Objectives

Reduce the waiting times of patients for an under-capacitated diagnostic resource.

- Qualitative Insights

Reasonable capacity protection reduces waiting times of prioritised patients, while excessive capacity protection results in a large number of unused slots. Additionally, reallocation of unused capacity can improve system efficiency, thereby improving waiting times of patients.

- Relevance to the Scheduling Policy

The H–L capacity protection mechanism proposed by Patrick and Puterman was directly implemented and formed the baseline model in Q4.

2.3 Chen and Robinson (2014)

- Essential Assumptions

Chen and Robinson (2014) considered appointment scheduling with both booked patients and same-day call-in patients, where more uncertain future high-priority demand is considered than

Patrick and Puterman's formulation.

- Main Objectives

Determine appointment sequences that minimise expected waiting costs.

- Qualitative Insights

Appointment scheduling fundamentally involves a trade-off between scheduling flexibility and capacity utilisation.

- Relevance to the Scheduling Policy

The study provides one of the key theoretical insights underpinning Q4: scheduling policy in priority-based appointment systems should balance response flexibility with resource utilisation.

2.4 Deglise-Hawkinson et al. (2018)

- Essential Assumptions

Deglise-Hawkinson et al. (2018) extended appointment scheduling to a broader healthcare capacity allocation problem involving more complicated system constraints and multiple patient groups with different priority levels and access requirements.

- Main Objectives

Determine appropriate capacity allocations that minimise weighted waiting-time measures across patient groups while satisfying the system constraints.

- Qualitative Insights

Priority-weighted capacity allocation differentiates access according to patient priority, which is able to improve overall system performance by allocating resources according to patient priority while maintaining equitable access across different patient groups.

- Relevance to the Scheduling Policy

This study provides theoretical support for the dynamic reservation mechanism developed in Q4 - adjusting appointment capacity for U patients and R patients according to the R workload.

3. Policies

3.1 Q2 - First Available Slot Policy (FAS)

- Structure

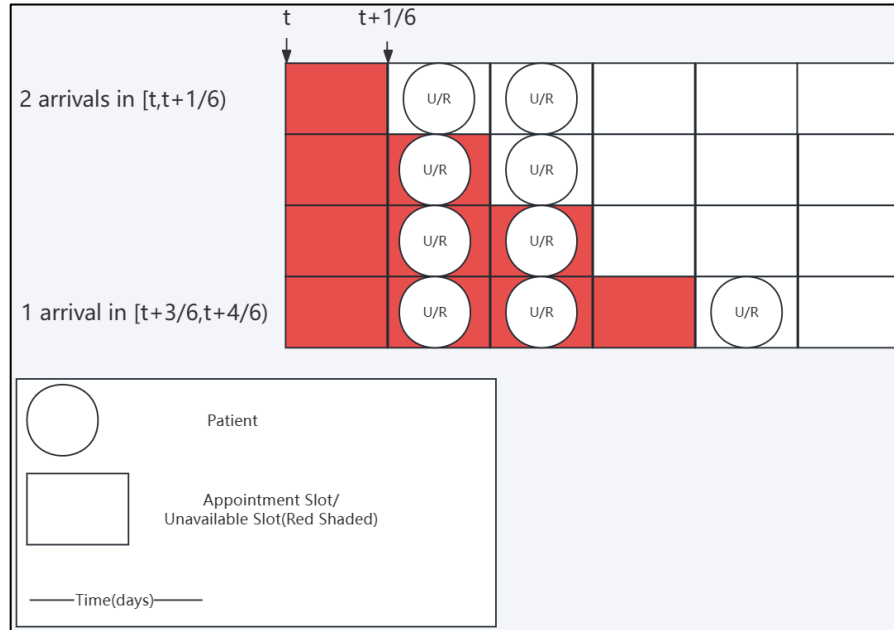


Figure 3.1.a - FAS Policy

Figure 3.1.a presents the FAS policy applied in Q2. The grid represents sequential appointment slots over time, where each column corresponds to a time slot of $1/6$ day. White circles indicate patients assigned to appointment slots, while red-shaded cells represent unavailable slots that are outdated.

Under FAS, each arrived patient is assigned to the earliest available appointment slot after arrival, regardless of patient type. When the earliest possible slot is occupied, the patient is shifted to the next available slot.

- Design Rationale

The FAS policy is provided by the guidance, serving as the baseline scheduling policy for subsequent experiments.

3.2 Q3 - U Workload Based Dynamic Delay Extension to FAS

- Structure

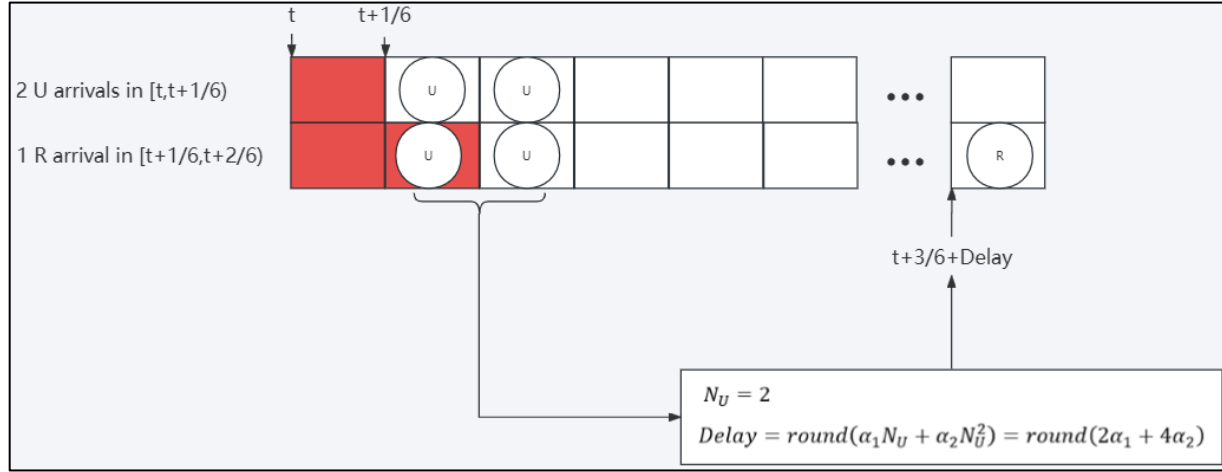


Figure 3.2.a - Urgent Workload Based Dynamic Delay Extension to FAS

Figure 3.2.a presents the Urgent Workload Based Dynamic Delay Extension to the FAS policy applied in Q3. (The notation follows the legend introduced in **Figure 3.1.a**.)

Under this policy, U patients are prioritised by dynamically delaying the appointment allocation of R patients according to the current U workload. When an R patient arrives, the earliest available slot is extended by a delay calculated from the observed number of U patients in the queue denoted as N_U . The delay is determined as

$$\text{round}(\alpha_1 N_U + \alpha_2 N_U^2) \quad (1)$$

where larger U workloads result in greater delay of R appointments, creating additional capacity for potential future U arrivals.

- Design Rationale

The policy structure is provided by the guidance to prioritise U patients while avoid excessively delay R patients.

3.3 Q4 - OCR Reallocation and R Workload-based Reservation Policy

- Structure

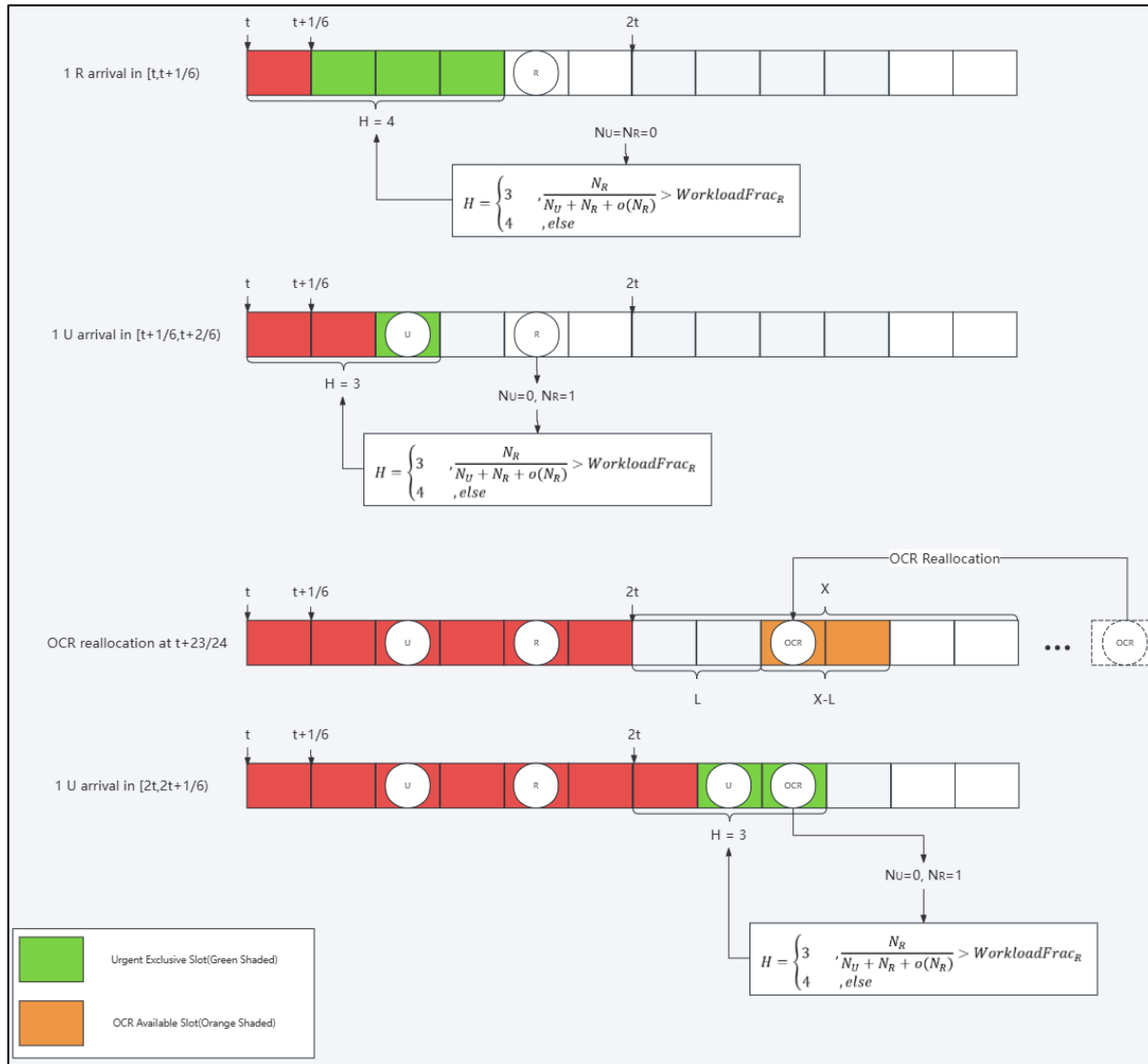


Figure 3.3.a - FAS with Dynamic Reservation Based on Routine Workload and OCR Reallocation

Figure 3.3.a presents the FAS with Dynamic Reservation and OCR Reallocation policy developed in Q4. (The notation follows the legend introduced in **Figure 3.1.a**, with green shaded slots representing capacity reserved exclusively for U patients and orange shaded slots available for OCR patients during reallocation.)

This policy consists of two stages:

Along the working hours, the number of reserved slots for U patients H is dynamically adjusted

according to the current workload composition. Specifically,

$$H = \begin{cases} H - 1, & \frac{N_R}{N_U + N_R} > \text{WorkloadFrac}_R, \\ H, & \text{otherwise,} \end{cases} \quad (2)$$

where N_U and N_R denote the numbers of U and R patients currently waiting. When the R workload exceeds the threshold WorkloadFrac_R , one reserved slot is released for R patient.

At the end of each day, if the following day contains $X > L$, the first $\min(X - L, N)$ OCR patients whose appointments occur earlier than reallocation slots are rescheduled in FIFO order to the earliest available appointment slots starting from $L + 1$, where X denotes the number of free slots of the following day.

- Design Rationale

The two-stage reservation framework proposed by Patrick and Puterman (2007) improves capacity utilisation while preserves capacity for future U arrivals. Building upon it, this study introduces a workload-based dynamic reservation mechanism to prevent excessive accumulation of waiting R patients, which is inspired by Deglise-Hawkinson et al. (2018) and findings in Q3.

4. Results

Unless otherwise stated, all subsequent simulation experiments are based on same initial configuration:

λ_U	3.69
λ_R	2.19
η	0.1
Warm-up Period	1000 th Served
Simulation End	50000 th Arrival
Simulation Replications	500

4.1 Q2 Results and Relevant Insights

4.1.1 Simulation Performance

Measure	Mean	Half-width of 95% CI	Relative Error
μ_U	4.126	0.113	2.739%
μ_R	4.124	0.113	2.740%
μ	4.126	0.113	2.739%

Table 4.1.1a - Performance under FAS Policy

Table 4.1.1.a presents the results under FAS policy:

- 1) μ is estimated as **4.126** days, while μ_U and μ_R are almost identical (**4.126** and **4.124**, respectively), verifying that FAS does not distinguish between patient types, which is consistent with the conclusion in **Section 4** of the coursework brief.
- 2) A total of **500** simulation replications was sufficient to ensure that the relative errors of all estimators were below **5%**, thereby providing sufficiently precise estimates.

4.1.2 Warm-up Assessment

To evaluate whether the benchmark warm-up period of **1,000** served patients is appropriate, a single simulation run was analysed using a moving window of **100** consecutive patients, which exploited the similarity of queueing behaviour over short intervals to smooth short-term fluctuations and reduce the influence of outliers, thereby improving the interpretability of the resulting plot.

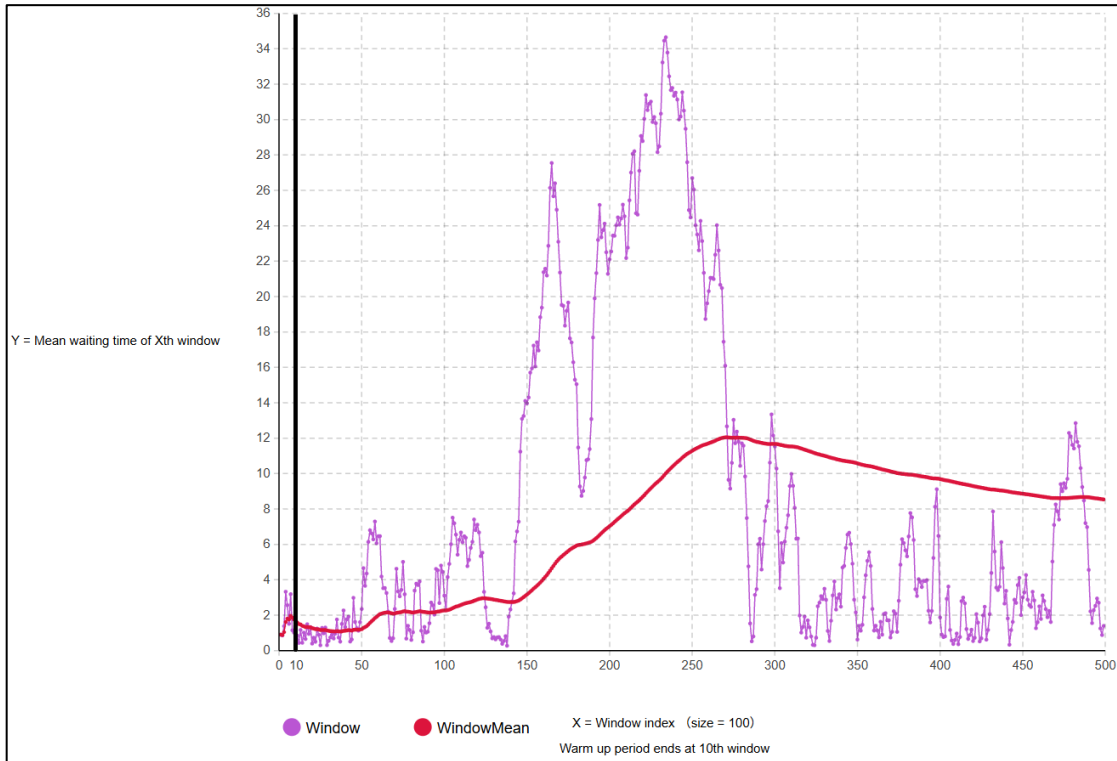


Figure 4.1.2.a - Average Waiting Time for Each Window

Figure 4.1.2.a presents the average waiting time for each window, with the warm-up period ending at the **10th** window:

- 1) During the initial windows, the system starts from an empty state, resulting in relatively low waiting times and limited variability, which is not fully representative of the long-run behaviour of the appointment system.
- 2) After warm-up period, the system reaches its typical congestion level and the waiting time begins to fluctuate around a stable range indicated by the red line.

Consequently, using **1,000** patients as the warm-up period is adequate for reducing transient bias introduced by the empty-system initialisation.

4.2 Q3 Results and Relevant Insights

4.2.1 Heuristic Parameter Search and Performance

Following the guidance, Q3 policy with $\alpha_1 = 0$ and a near-optimal α_2 is considered competitive, thus α_1 was fixed at **0** and search range is based on α_2 .

α_1	α_2	μ_U	μ_R	μ
0	0.8	1.937	12.79	5.978
0	0.7	1.958	11.417	5.478
0	0.64	1.984	10.808	5.269
0	0.62	1.993	10.536	5.175
0	0.6	1.998	10.300	5.088
0	0.58	2.024	10.102	5.030
0	0.56	2.005	9.582	4.829
0	0.5	2.063	9.176	4.710
0	0.4	2.197	8.320	4.477
0	0.3	2.362	7.220	4.170
0	0.2	2.725	6.494	4.129

Table 4.2.1.a – Optimal Parameter Search under Q3 Policy

Table 4.2.1.a present results under Q3 policy: ($\alpha_1 = \mathbf{0}$, $\alpha_2 = \mathbf{0.56}$) is the best configuration within the experimental search range, achieving $\mu_U = \mathbf{2.005}$ and $\mu = \mathbf{4.829}$, which satisfies the constraints ($\mu_U^\pi < \mu^F, \mu^\pi \leq \mathbf{5}$). Compared with the FAS policy ($\mu_U = \mathbf{4.126}$, $\mu = \mathbf{4.126}$), μ_U decreases by approximately **51.406%**, while μ increases about **17.038%**.

4.2.2 Relationship between α_2 and the Performance Measures

To further investigate the effect of the α_2 in performance measures, linear regression was performed using the simulation results presented in **Table 4.2.1.a**.

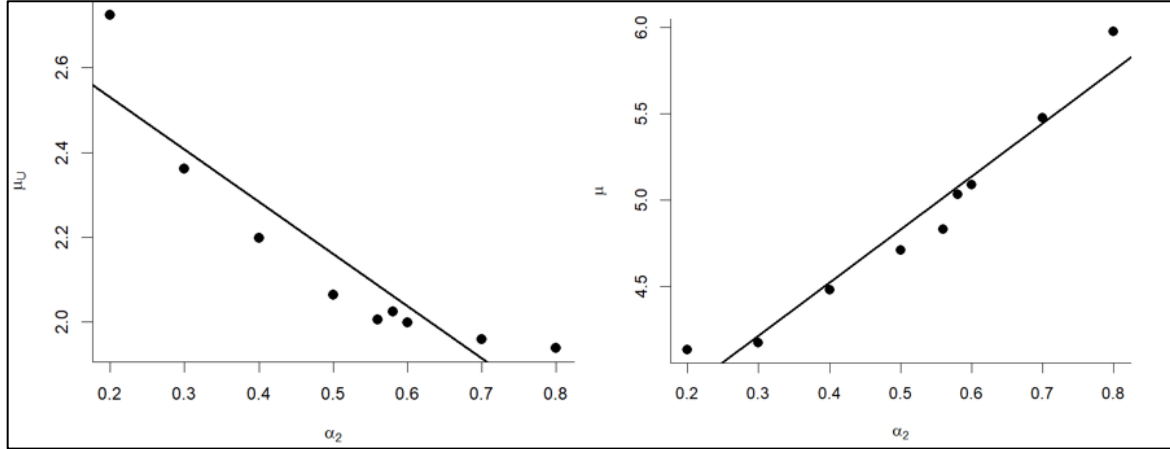


Figure 4.2.2.a - Side-by-side Plots of α_2 vs μ_U and μ

Figure 4.2.2.a demonstrates the relationship between α_2 and μ_U and μ . The fitted regression models provide a good fit to the simulation data, with R^2 values of **0.829** and **0.945**, and adjusted R^2 values of **0.805** and **0.937**, respectively:

$$\mu_U(\alpha_2) = 2.776 - 1.232\alpha_2 \quad (3)$$

$$\mu(\alpha_2) = 3.288 + 3.081\alpha_2 \quad (4)$$

The regression results reveal important findings:

- 1) Increasing α_2 produces two opposite monotonic trends: μ_U decreases approximately linearly, whereas μ increases approximately linearly. In the Q3 policy, this trade-off arises because appointments for R patients are intentionally postponed to preserve capacity for U patients, leading to the accumulation of waiting time among R patients, which could be illustrated intuitively by the following equations:

As a simplifying approximation, μ is assumed to satisfy

$$(\lambda_U + \lambda_R)\mu \approx \lambda_U\mu_U + \lambda_R\mu_R = 3.69\mu_U + 2.19\mu_R \quad (5)$$

Combining Equations (3) and (4) yields the following approximate linear relationship between μ and μ_U under Q3 policy

$$\mu \approx -2.501\mu_U + 10.230 \quad (6)$$

Substituting Equation (6) into Equation (5)

$$-14.706\mu_U + 60.152 \approx 3.69\mu_U + 2.19\mu_R \quad (7)$$

By manipulating Equation (7), could obtain

$$\mu_R \approx \frac{60.152 - 18.396\mu_U}{2.19} \quad (8)$$

For convenience, the trade-off revealed by Equation (8) is hereafter referred to as the **U-R trade-off**.

- 2) This observation is consistent with the findings of Deglise-Hawkinson et al. (2018), who similarly reported that priority-weighted capacity allocation differentiates access according to patient priority. Both Deglise-Hawkinson's finding and the fitted models suggest that all appointment systems with capacity protection exhibit an inherent U-R trade-off.
- 3) Accordingly, $\alpha_2 = 0.56$ provides a feasible balance, which is considered competitive.

4.3 Q4 Results and Relevant Insights

4.3.1 Evaluation of Patrick and Puterman Policy

The first experiment investigated the influence of the reservation horizon H under the Patrick and Puterman policy recommended in the guidance, to examine whether the policy behaves as expected. Experimental configuration was as follows:

- 1) L was treated as a control variable and fixed at **0**.
- 2) H was varied from **5** to **1** following Patrick and Puterman's definition.

H	L	μ_U	μ_R	μ
5	0	0.479	1182.014	269.476
4	0	1.895	56.872	22.256
3	0	4.188	4.782	4.409
2	0	4.034	4.254	4.116
1	0	4.039	4.109	4.065

Table 4.3.1.a - Performance Measures under Different H

As demonstrated in **Table 4.3.1.a**, increasing H from 3 to 4 substantially reduced μ_U from **4.188** to **1.895**, with μ_R increasing sharply from **4.782** to **56.872** and μ rising to an unacceptable **22.256**.

In order to further optimize, the second experiment investigates how different values of L affect μ , by evaluating all feasible L values (from **0** to $H - 1$). Experimental configuration was as follows:

- 1) Appointment slot utilisation was introduced as an additional performance measure to assess whether OCR reallocations could significantly improve overall slot utilisation, which is measured as

$$utilisation = \frac{N_{booked}(t)}{N_{total}(t)} \quad (9)$$

where $N_{booked}(t)$ is the number of appointment slots booked up to time t , and $N_{total}(t)$ is the total number of appointment slots generated up to time t .

- 2) $H = 3$ and $H = 4$ were selected as the starting configurations, as the optimal policy consistent with the Q4 objective is expected to lie between these two reservation levels

according to previous experiment.

- 3) For each value of H, L was varied from **0** to **H – 1** to evaluate all feasible combinations.

H	L	μ_U	μ_R	μ	utilisation
4	0	1.895	56.872	22.256	0.973
4	1	1.797	67.459	26.093	0.972
4	2	1.757	128.308	47.995	0.967
4	3	1.69	286.295	102.885	0.954
3	0	4.188	4.782	4.409	0.978
3	1	4.126	4.725	4.349	0.978
3	2	4.188	4.792	4.413	0.978

Table 4.3.1.b - Performance Measures under Different L

Table 4.3.1.a and **Table 4.3.1.b** reveal several important findings:

- 1) Adjusting H changes the level of priority protection and produces a similar U-R trade-off discussed in **Section 4.2.2**.
- 2) Across all parameter settings, appointment-slot utilisation remained above **95%**, which is consistent with the theoretical demand-to-capacity ratio of

$$\frac{(\lambda_U + \lambda_R)}{6} = \frac{5.88}{6} \approx 98\% \quad (10)$$

This suggests that the appointment system operates close to its capacity limit, leaving little spare capacity for OCR reallocations, which supports the utilisation–flexibility trade-off discussed by Chen and Robinson (2014). Therefore, only limited opportunities existed for OCR reallocations throughout the planning horizon, regardless of the choice of L.

- 3) Meanwhile, the effect of L varies from different H. Under the aggressive reservation policy ($H = 4$), increasing L from **0** to **3** increased μ_R dramatically from **56.872** to **286.295**. Consequently, μ increased from **22.256** to **102.885**. One possible explanation is that, under $H = 4$, the reserved capacity slightly exceeds the average daily U arrival rate ($\lambda_U = 3.69 < 4$), potentially creating vacant slots. However, increasing L imposes a stricter limitation on the availability of vacant slots for OCR reallocation, thereby increasing μ_R when R patients are already heavily delayed.
- 4) Conversely, a markedly different pattern is observed when $H = 3$, where varying L

produces only negligible changes in all performance measures. One possible explanation is that the reserved capacity does not consistently exceed the average daily arrival rate of U patients ($\lambda_U = 3.69 > 3$), making the reserved slots filled immediately, thereby leaving very few vacant slots for subsequent OCR reallocations. As the reallocation mechanism is rarely activated, modifying L has virtually no influence on system performance.

All the findings suggest that the better policy lie between $H = 3$ and $H = 4$ while L fixed 0. Consequently, this motivates the introduction of a more adaptive reservation policy in the following experiment to further optimize.

4.3.2 OCR Reallocation and R Workload-Based Reservation Policy Performance

The Q4 policy in the third experiment introduces a new parameter, WorkloadFracR, which dynamically limits reserved capacity (and OCR reallocations by restricting the upper limit of L) according to the current R workload fraction on Patrick and Puterman Policy (See Equation (2)), inspired by the U-R trade-off found in **Section 4.2.2**. Experimental configuration was set as follow:

- 1) Based on the findings of the previous experiment, $L = 0$ was fixed to maximize OCR reallocation opportunities.
- 2) Meanwhile, the reservation level was initialised at $H = 4$, since the Q4 policy dynamically switches the reservation level between $H = 4$ and $H = 3$ according to the R workload and any other H would be inconsistent with the policy design.
- 3) WorkloadFracR was varied from **1.0** to **0.6**, where ($H = 4$, WorkloadFracR = **1.0**) serves as the benchmark (R workload ratio never exceed **1**), making this configuration equivalent to the original Patrick and Puterman policy.

H	L	WorkloadFracR	μ_U	μ_R	μ
4	0	1	1.887	57.975	22.663
4	0	0.98	1.889	13.574	6.238
4	0	0.96	1.939	11.315	5.429
4	0	0.94	1.993	10.037	4.998
4	0	0.92	2.087	9.657	4.907
4	0	0.9	2.179	9.274	4.822
4	0	0.8	2.58	8.183	4.667
4	0	0.7	3.031	6.943	4.488
4	0	0.6	3.528	6.199	4.523

Table 4.3.2.a - Performance Measures under Different R Workload Thresholds

Table 4.3.2.a presents that as the workload threshold decreases from **1.0**, the Q4 policy becomes increasingly responsive to routine-patient congestion by switching from $H = 4$ to $H = 3$. Among all values, $\text{WorkloadFracR} = 0.94$ provides the most favourable balance between μ_U (**1.993**) and μ_R (**10.037**), which satisfies the constraints (μ_U improved, $\mu < 5$).

Important findings from the results are as follows:

- 1) Compared with the best policy identified in Q3 ($\mu_U = 2.005$, $\mu_R = 9.582$, $\mu = 4.829$), the best Q4 policy decreases μ_U by **0.6%**, but increases μ_R by **4.7%**, thereby increases μ by **3.5%**. Therefore, the marginal improvement in μ_U time was outweighed by the deterioration in μ , indicating that, under current system configuration, the best Q4 policy did not outperform the best Q3 policy.
- 2) The best Q4 policy optimizes μ_U mainly by achieving a U-R trade-off rather than improving utilisation, since as discussed in **Section 4.3.1**, the system already operates under high utilisation.

5. Conclusions

The main conclusions are as follows:

- 1) The simulation experiments validated that all policies (Q2–Q4) satisfy the predefined constraints while achieving the intended optimisation objectives.
- 2) Analysis in Q3 revealed an inherent U–R trade-off that exists in appointment systems with capacity protection and quantified it using Equation (8), which is consistent with the qualitative findings reported in the literature.
- 3) Analysis in Q4 revealed that, under the current configuration, the appointment system operated under high utilisation, leaving limited opportunities for OCR reallocations to further improve utilisation.
- 4) Motivated by the U–R trade-off identified in Q3, a workload-based dynamic reservation policy was proposed in Q4 to balance the competing waiting times of the two patient groups in order to minimising μ_U subject to the constraint $\mu < 5$.

Several limitations should also be acknowledged:

- 1) Selected parameters in Q3 and Q4 were obtained through heuristic search rather than systematic optimisation, where parameter configurations with finer resolutions were not investigated. Consequently, the optimisation is limited by the precision of the parameter search.
- 2) The Q4 policy does not further increase utilisation, though improves μ_U under the required constraints through dynamic capacity reallocation. Consequently, the policy cannot further optimise waiting times for both patient groups simultaneously.

These conclusions and limitations provide directions for future research:

- 1) More systematic optimisation approaches within a carefully defined parameter space, could be applied to identify more refined policy parameters and potentially achieve further performance improvements.
- 2) Moreover, investigating additional mechanisms to enhance utilisation under the Q4 policy may provide opportunities to further reduce waiting times for both patient groups. Specifically, predicting future idle capacity may create more opportunities for OCR reallocations, thereby significantly improving utilisation.

- 3) Incorporating more realistic operational characteristics, like patient preferences and random cancellations, would also help improve the robustness and generalizability of the policies.

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